

Reflection Essay Task 02

When I first read the assignment, projective geometry felt like one of those topics that is elegant on the whiteboard but difficult to appreciate fully without actually applying it. Concepts such as vanishing points, cross-ratio invariance, perspective projection, and distance estimation are mathematically beautiful, but they can also feel abstract when viewed only through formulas. This assignment changed that experience for me. By collecting my own images, running object detection, annotating points, estimating a vanishing point, and finally computing real-world distances, I was able to connect the mathematical concepts discussed in class to a tangible real-world scenario.

I decided to create my own experimental setup. I selected a long straight section of the Mumbai promenade because it naturally contained several useful geometric features:

- A clearly visible ground plane.
- Long straight tile seams.
- Strong perspective effects.
- Distance markers at 900 m and 1000 m.
- Sufficient depth in the scene.
- A safe and accessible environment for image collection.



To better document the experiment, I captured both a first-person view (FPV) image and a third-person view (TPV) image.



The FPV image became the primary image used for computation because it contained strong perspective cues and a clearly visible subject. The TPV image served as a useful reference for understanding the overall scene setup and validating the physical arrangement of the experiment.

One aspect I particularly enjoyed was that the experiment did not begin with code. It began with observation.

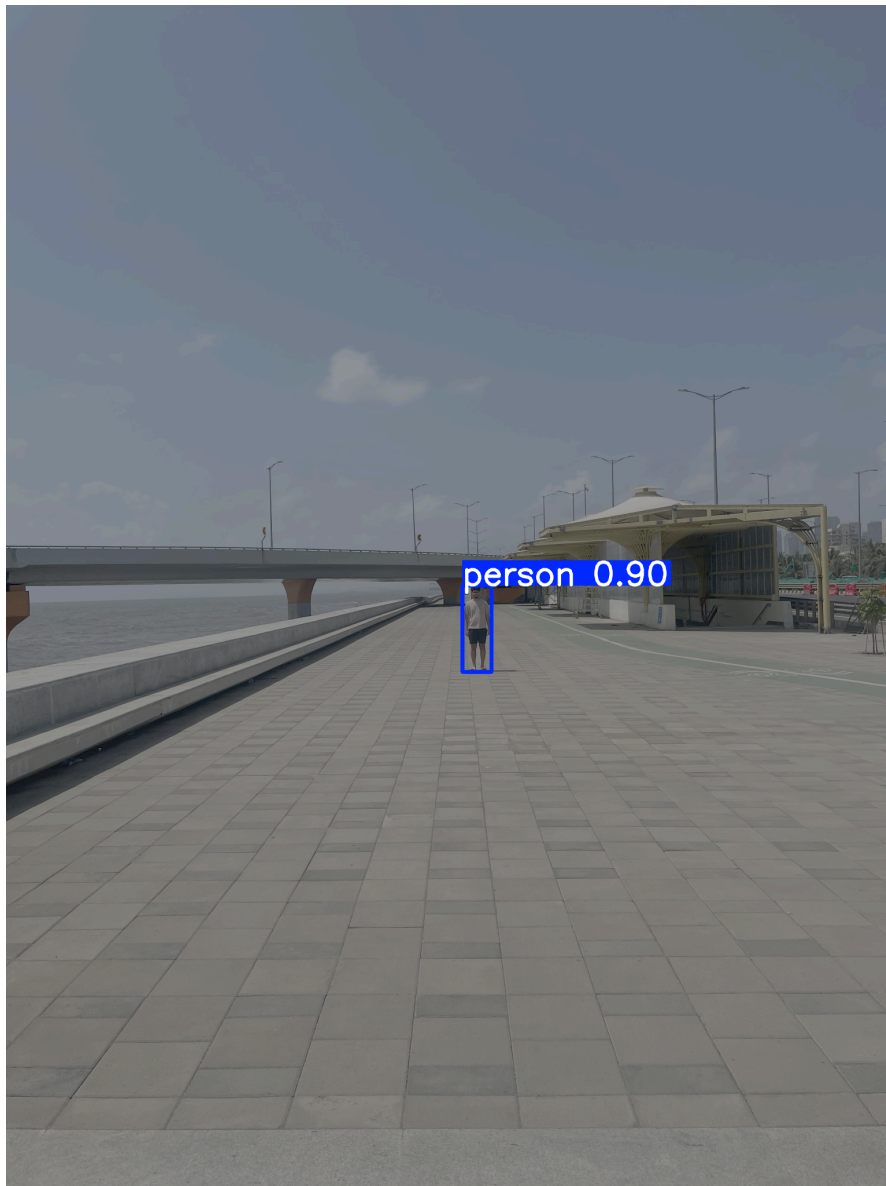
Before writing a single line of Python, I spent time examining the scene and thinking about questions such as:

- Which lines in the image are actually parallel in the real world?
- Which point on a person should represent their location?

- How should a known distance be selected?
- Which image would produce the most reliable geometric estimate?
- How can annotation errors be reduced?

This process helped me appreciate that successful computer vision often begins with understanding the scene rather than immediately applying algorithms.

The next stage involved object detection using YOLOv8. I used YOLO to detect the person present in the scene and obtain a bounding box. The detector successfully identified the person with high confidence.



However, obtaining a bounding box was only the beginning.

The more interesting question was determining which point from the bounding box should be used for projective geometry calculations.

After considering different possibilities, I selected the bottom-center point of the bounding box because it best approximates the physical location where the person touches the ground.

This decision may appear simple at first, but it reinforced an important lesson:

- Computer vision outputs often require interpretation.
- Geometric reasoning is necessary even after detection succeeds.
- The "correct" point depends on the problem being solved.
- Understanding the scene is just as important as understanding the model.

The combination of object detection and geometry was one of the most rewarding aspects of the assignment.

YOLO answered the question:

"What object is present in the image?"

Projective geometry answered the question:

"Where is that object located in the real world?"

Seeing these two ideas work together provided a broader understanding of how different computer vision techniques complement one another.

After detecting the person, I moved on to vanishing point estimation.

This was perhaps the most interesting stage of the entire workflow.

Instead of estimating the vanishing point visually, I used the tile seams visible on the promenade. Since these seams are parallel in the real world, their image projections must intersect at a common vanishing point.

I created small Python scripts using OpenCV to collect exact pixel coordinates and compute the line intersection programmatically.

This part of the assignment taught me several practical lessons:

- Precise coordinates matter.
- Small annotation differences can affect downstream calculations.
- Automation improves reproducibility.

- Geometry becomes much easier to trust when computations are performed programmatically.

The vanishing point calculation transformed what initially seemed like an abstract classroom concept into something concrete and measurable.

Another learning experience came from constructing the projective line.

Initially, it is tempting to draw a line manually and assume the selected points are sufficiently accurate. However, I learned that mathematical consistency is important.

To ensure collinearity, I projected the selected marker locations onto the line connecting the person's ground-contact point and the vanishing point.

This approach helped guarantee that:

- The selected points satisfied the geometric assumptions.
- The cross-ratio calculation remained valid.
- The methodology could be reproduced consistently.

Once all four points had been identified, the final stage was distance estimation using cross-ratio invariance.

This was the moment where everything came together.

Throughout the lectures, the cross-ratio was presented as a quantity that remains invariant under perspective projection. While I understood the derivation mathematically, the practical significance became much clearer during this experiment.

The fact that:

- Distances in the image are distorted,
- Objects shrink with depth,
- Perspective changes apparent spacing,

and yet a meaningful geometric relationship survives these distortions is remarkably elegant.

Applying the cross-ratio to my own collected data and obtaining a real-world distance estimate was probably the most satisfying part of the assignment.

Another aspect I appreciated was the iterative nature of the workflow.

The final result was not produced by a single algorithm. Instead, it emerged from multiple connected stages:

- Scene selection.
- Data collection.
- Object detection.
- Coordinate annotation.
- Vanishing point estimation.
- Projective line construction.
- Cross-ratio computation.
- Result verification.

Each stage contributed to the final estimate.

This reinforced an important lesson about computer vision projects: success often depends more on the quality of the overall pipeline than on any single component.

Beyond the technical content, the assignment also changed how I look at everyday scenes.

After spending time identifying vanishing points and perspective lines, I began noticing them almost everywhere:

- Roads.
- Footpaths.
- Railway tracks.
- Building corridors.
- Long hallways.
- Urban streetscapes.

Scenes that previously appeared ordinary started revealing underlying geometric structure.

This was one of the most enjoyable outcomes of the assignment because it demonstrated how concepts learned in class can influence how we observe the world.

Some of my key takeaways from the project are:

- Projective geometry provides powerful tools for extracting metric information from a single image.
- Object detection and geometry complement each other effectively.
- Careful scene selection can significantly improve results.
- Reliable annotations are critical for geometric computations.
- Small utility scripts can greatly improve accuracy and reproducibility.

- Mathematical concepts become much easier to understand when applied to real data.
- Verification against physical intuition remains an important part of the workflow.

Overall, this assignment was one of the most engaging computer vision exercises I have completed. It combined theory, experimentation, programming, observation, and mathematical reasoning in a way that felt both challenging and rewarding.

Multiple Images tried and best most aligned one picked!

